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ASSIGNMENT 10

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The Two-Group Normal Model: Two Parameterizations

Assignment 10 asks that the entire analysis of the model worked in lecture—one control group and one treated group—be carried out twice, under two parameterizations, and that the matrix and the non-matrix routes be shown to agree at every step. This preamble fixes the notation once so that the eleven parts can refer back to it.

The data. There are two independent groups. The control group (group 1) contributes observations $Y_{11}, \ldots, Y_{1n_1}$ and the treated group (group 2) contributes $Y_{21}, \ldots, Y_{2n_2}$, with $n = n_1 + n_2$ observations in all. As the assignment stipulates, the "error terms" are independent, normal, and share a common variance: writing μ_g for the expectation in group g,

$$Y_{gi} \ = \ \mu_g + \varepsilon_{gi}, \qquad \varepsilon_{gi} \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

so that $Y_{1i} \sim N(\mu_1, \sigma^2)$ and $Y_{2j} \sim N(\mu_2, \sigma^2)$, all mutually independent. It is convenient to record the group sums and means

$$S_1 = \sum_{i=1}^{n_1} Y_{1i} = n_1 \bar{Y}_1, \qquad S_2 = \sum_{j=1}^{n_2} Y_{2j} = n_2 \bar{Y}_2, \qquad S = S_1 + S_2.$$

Parameterization A (group means). Take the two group expectations themselves as the parameters:

$$\mathrm{E}\{Y_{1i}\} = a_1 \quad \text{(control)}, \qquad \mathrm{E}\{Y_{2j}\} = a_2 \quad \text{(treated)}.$$

Parameterization B (the lecture's regression coding). Take an intercept and an increment: the first group's expectation is b_0, and the treated group is displaced from it by b_1,[1]

$$\mathrm{E}\{Y_{1i}\} = b_0, \qquad \mathrm{E}\{Y_{2j}\} = b_0 + b_1.$$

The two parameterizations describe the same family of distributions and are related by the invertible linear change of variable

$$b_0 = a_1, \qquad b_1 = a_2 - a_1, \qquad \text{equivalently} \qquad a_1 = b_0, \quad a_2 = b_0 + b_1.$$

Parameterization A reads off each group mean directly; parameterization B isolates the treatment effect $b_1 = a_2 - a_1$, which is usually the quantity of interest.

Part 1

Write out the likelihood and the log likelihood without using matrices.

Solution:

Independence lets the joint density factor into a product of the n normal densities, one per observation. Grouping the two samples,

$$L(a_1, a_2, \sigma^2) = \prod_{i=1}^{n_1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{ - \frac{(Y_{1i} - a_1)^2}{2\sigma^2} \Big\} \prod_{j=1}^{n_2} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{ - \frac{(Y_{2j} - a_2)^2}{2\sigma^2} \Big\},$$

[1] The assignment writes "the expectation in the first group is b_0 and the expectation in the second group is b_1." This is the lecture's dummy-variable coding, in which b_1 is the *regression coefficient* on the treatment indicator—the slope—so that the treated-group expectation is $b_0 + b_1$, not b_1 itself. Reading it the other way (b_1 the treated mean outright) would make the two parameterizations literally identical and empty the exercise of content; the whole point of the contrast is that b_1 measures the treatment *effect* $a_2 - a_1$.

which collects into

$$\boxed{L(a_1, a_2, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\Big\{ -\frac{1}{2\sigma^2}\Big[\sum_{i=1}^{n_1}(Y_{1i}-a_1)^2 + \sum_{j=1}^{n_2}(Y_{2j}-a_2)^2\Big]\Big\}.}$$

Taking logs turns the product into a sum and the exponential into its argument:

$$\boxed{\ell(a_1, a_2, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\Big[\sum_{i=1}^{n_1}(Y_{1i}-a_1)^2 + \sum_{j=1}^{n_2}(Y_{2j}-a_2)^2\Big].}$$

For parameterization B substitute $a_1 \mapsto b_0$ in the control group and $a_2 \mapsto b_0 + b_1$ in the treated group:

$$\ell(b_0, b_1, \sigma^2) = -\frac{n}{2}\ln\big(2\pi\sigma^2\big) - \frac{1}{2\sigma^2}\Big[\sum_{i=1}^{n_1}(Y_{1i}-b_0)^2 + \sum_{j=1}^{n_2}(Y_{2j}-b_0-b_1)^2\Big].$$

Only the way the sum of squares is written down has changed; the number in the bracket is the same for both parameterizations at corresponding parameter values, because $Y_{2j} - a_2 = Y_{2j} - (b_0 + b_1)$ term by term. That the log likelihood depends on the data only through the two within-group sums of squares is the first hint of the sufficiency that later parts exploit. ■

Part 2

Write out the likelihood and the log likelihood using matrix notation. Be sure to say what is your X matrix.

SOLUTION:

Stack the observations, control group first, into the $n \times 1$ vector $\mathbf{Y} = (Y_{11}, \dots, Y_{1n_1}, Y_{21}, \dots, Y_{2n_2})^\top$. The model is the linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N_n(\mathbf{0},\ \sigma^2 \mathbf{I}_n),$$

so $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$. Because the covariance is the scalar matrix $\sigma^2\mathbf{I}_n$, its determinant is $(\sigma^2)^n$ and the quadratic form in the exponent is $\sigma^{-2}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$. Hence

$$\boxed{L(\boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\Big\{ -\frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \Big\},}$$

$$\boxed{\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln\big(2\pi\sigma^2\big) - \frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).}$$

Expanding the quadratic form reproduces exactly the bracket of Part 1, since $(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \sum_{\text{all obs}}(\text{observation} - \text{its fitted mean})^2$.

The design matrix. What differs between the two parameterizations is only $\mathbf{X}$ and $\boldsymbol{\beta}$; both are $n \times 2$ and 2×1 respectively, with the first n_1 rows belonging to the control group.

For parameterization A, $\boldsymbol{\beta} = (a_1, a_2)^\top$ and the columns are the two group indicators (*cell-means* coding): row g has a 1 in the column of its own group and a 0 in the other,

$$\mathbf{X}_A = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0}_{n_1} \\ \mathbf{0}_{n_2} & \mathbf{1}_{n_2} \end{pmatrix} \quad \text{so that} \quad \mathbf{X}_A \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 \mathbf{1}_{n_1} \\ a_2 \mathbf{1}_{n_2} \end{pmatrix}.$$

For parameterization B, $\boldsymbol{\beta} = (b_0, b_1)^\top$ and the first column is an intercept of all ones while the second column is the treatment indicator, 1 for the treated rows and 0 for the controls (*reference* or dummy coding):

$$\mathbf{X}_B = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0}_{n_1} \\ \mathbf{1}_{n_2} & \mathbf{1}_{n_2} \end{pmatrix} \quad \text{so that} \quad \mathbf{X}_B \begin{pmatrix} b_0 \\ b_1 \end{pmatrix} = \begin{pmatrix} b_0 \mathbf{1}_{n_1} \\ (b_0 + b_1) \mathbf{1}_{n_2} \end{pmatrix}.$$

Here $\mathbf{1}_m$ and $\mathbf{0}_m$ are the $m \times 1$ all-ones and all-zeros vectors. The two design matrices have the same column space—each spans all vectors constant within groups—which is why they fit the identical model; they differ only in the basis chosen for that space. Indeed $\mathbf{X}_B = \mathbf{X}_A T$ with $T = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, the same invertible reparameterization $b_0 = a_1,\ b_1 = a_2 - a_1$ recorded in the preamble. ■

Part 3

Write out the score equations for computing the MLE and check that you get the same result whether or not you use the matrix notation.

SOLUTION:

Without matrices. Differentiate the Part 1 log likelihood. For parameterization A the coefficient scores are

$$\frac{\partial \ell}{\partial a_1} = \frac{1}{\sigma^2}\sum_{i=1}^{n_1}(Y_{1i} - a_1) \stackrel{!}{=} 0, \qquad \frac{\partial \ell}{\partial a_2} = \frac{1}{\sigma^2}\sum_{j=1}^{n_2}(Y_{2j} - a_2) \stackrel{!}{=} 0,$$

i.e. each group's residuals must sum to zero, giving $\sum_i (Y_{1i} - a_1) = 0$ and $\sum_j (Y_{2j} - a_2) = 0$. For parameterization B, the chain rule (both group means depend on b_0, only the treated mean on b_1) gives

$$\frac{\partial \ell}{\partial b_0} = \frac{1}{\sigma^2}\Big[\sum_{i=1}^{n_1}(Y_{1i} - b_0) + \sum_{j=1}^{n_2}(Y_{2j} - b_0 - b_1)\Big] \stackrel{!}{=} 0, \qquad \frac{\partial \ell}{\partial b_1} = \frac{1}{\sigma^2}\sum_{j=1}^{n_2}(Y_{2j} - b_0 - b_1) \stackrel{!}{=} 0.$$

The variance score is the same in either coding,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\Big[\sum_i (Y_{1i} - \hat{\mu}_1)^2 + \sum_j (Y_{2j} - \hat{\mu}_2)^2\Big] \stackrel{!}{=} 0.$$

With matrices. Using $\partial\big[(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\big]/\partial\boldsymbol{\beta} = -2\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$,

$$\boxed{\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \frac{1}{\sigma^2}\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0} \iff \mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{Y}} \qquad \text{(the normal equations)},$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = 0.$$

They agree. The vector score $\mathbf{X}^\top(\mathbf{Y}-\mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$ says the residual vector is orthogonal to each column of $\mathbf{X}$, i.e. each column's dot product with the residuals is zero. For $\mathbf{X}_A$ the columns are the group indicators, so the two components read $\sum_i(Y_{1i}-a_1) = 0$ and $\sum_j(Y_{2j} - a_2) = 0$—the parameterization-A scalar equations exactly. For $\mathbf{X}_B$ the first column is all ones, giving the *total* residual sum $\sum_i(Y_{1i} - b_0) + \sum_j(Y_{2j} - b_0 - b_1) = 0$, and the second column is the treated indicator, giving $\sum_j(Y_{2j} - b_0 - b_1) = 0$—the parameterization-B scalar equations exactly. The matrix score is thus just the pair of scalar scores bundled into a vector, and the variance score is identical because the quadratic form equals the bracketed sum of squares. ■

Part 4

Find the MLEs both using matrix notation and not.

Solution:

Without matrices. The coefficient scores solve immediately. In parameterization A each group residual sums to zero at the group mean,

$$\boxed{\hat{a}_1 = \bar{Y}_1 = \frac{S_1}{n_1}, \qquad \hat{a}_2 = \bar{Y}_2 = \frac{S_2}{n_2}.}$$

In parameterization B the second equation gives $\hat{b}_0 + \hat{b}_1 = \bar{Y}_2$, and substituting into the first gives $\hat{b}_0 = \bar{Y}_1$, hence

$$\boxed{\hat{b}_0 = \bar{Y}_1, \qquad \hat{b}_1 = \bar{Y}_2 - \bar{Y}_1.}$$

These obey $\hat{b}_0 = \hat{a}_1$ and $\hat{b}_1 = \hat{a}_2 - \hat{a}_1$: the estimators transform the same way the parameters do, which is the invariance of the MLE under reparameterization. Solving the variance score with $\hat{\mu}_1 = \bar{Y}_1,\ \hat{\mu}_2 = \bar{Y}_2$ gives the common

$$\boxed{\hat{\sigma}^2 = \frac{1}{n}\Big[\sum_{i=1}^{n_1}(Y_{1i} - \bar{Y}_1)^2 + \sum_{j=1}^{n_2}(Y_{2j} - \bar{Y}_2)^2\Big] = \frac{\text{RSS}}{n},}$$

identical in both codings because the fitted means—and therefore the residuals—are the same.

With matrices. The normal equations give $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ whenever $\mathbf{X}$ has full column rank (here rank 2, provided $n_1, n_2 \geq 1$). For parameterization A,

$$\mathbf{X}_A^\top \mathbf{X}_A = \begin{pmatrix} n_1 & 0 \\ 0 & n_2 \end{pmatrix}, \quad \mathbf{X}_A^\top \mathbf{Y} = \begin{pmatrix} S_1 \\ S_2 \end{pmatrix}, \quad \Longrightarrow \quad \hat{\boldsymbol{\beta}} = \begin{pmatrix} S_1/n_1 \\ S_2/n_2 \end{pmatrix} = \begin{pmatrix} \bar{Y}_1 \\ \bar{Y}_2 \end{pmatrix},$$

matching $\hat{a}_1, \hat{a}_2$. For parameterization B,

$$\mathbf{X}_B^\top \mathbf{X}_B = \begin{pmatrix} n & n_2 \\ n_2 & n_2 \end{pmatrix}, \qquad (\mathbf{X}_B^\top \mathbf{X}_B)^{-1} = \frac{1}{n_1 n_2} \begin{pmatrix} n_2 & -n_2 \\ -n_2 & n \end{pmatrix}, \qquad \mathbf{X}_B^\top \mathbf{Y} = \begin{pmatrix} S \\ S_2 \end{pmatrix},$$

using $\det\big(\mathbf{X}_B^\top \mathbf{X}_B\big) = n\,n_2 - n_2^2 = n_2(n - n_2) = n_1 n_2$. Multiplying out,

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \frac{1}{n_1 n_2} \begin{pmatrix} n_2 S - n_2 S_2 \\ -n_2 S + n S_2 \end{pmatrix} = \frac{1}{n_1 n_2} \begin{pmatrix} n_2 S_1 \\ n_1 S_2 - n_2 S_1 \end{pmatrix} \\ &= \begin{pmatrix} S_1/n_1 \\ S_2/n_2 - S_1/n_1 \end{pmatrix} = \begin{pmatrix} \bar{Y}_1 \\ \bar{Y}_2 - \bar{Y}_1 \end{pmatrix}, \end{aligned}$$

where $n_2 S - n_2 S_2 = n_2 S_1$ and $-n_2 S + n S_2 = -n_2 S_1 + n_1 S_2$ follow from $S = S_1 + S_2$ and $n = n_1 + n_2$. This matches $\hat{b}_0, \hat{b}_1$. Finally $\hat{\sigma}^2 = \frac{1}{n}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathrm{RSS}/n$ as before, since $\mathbf{X}\hat{\boldsymbol{\beta}}$ is the vector of group means either way.

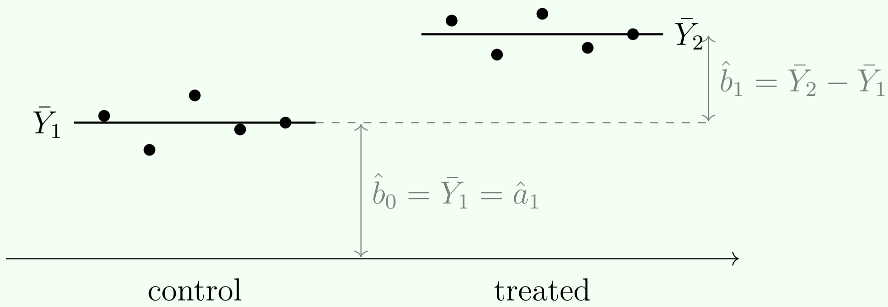

Figure 1. THE MLE IS THE PAIR OF GROUP MEANS. Least squares (equivalently, maximum likelihood under normality) sets each fitted value to its group average. Parameterization A reports the two levels $\bar{Y}_1, \bar{Y}_2$; parameterization B reports the control level $\hat{b}_0 = \bar{Y}_1$ and the vertical jump $\hat{b}_1 = \bar{Y}_2 - \bar{Y}_1$ to the treated level—the same fit, read two ways.

■

Part 5

Show that the MLEs for the regression coefficients are unbiased.

SOLUTION:

Matrix argument (covers both parameterizations at once). The estimator is linear in $\mathbf{Y}$, and $\mathrm{E}\{\mathbf{Y}\} = \mathbf{X}\boldsymbol{\beta}$, so

$$\mathrm{E}\{\hat{\boldsymbol{\beta}}\} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\,\mathrm{E}\{\mathbf{Y}\} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \boldsymbol{\beta}.$$

The matrix $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}$ collapses to the identity, leaving $\boldsymbol{\beta}$ untouched, for *whichever* design matrix is used. Hence $\mathrm{E}\{\hat{\boldsymbol{\beta}}\} = \boldsymbol{\beta}$ for every $\boldsymbol{\beta}$ and every σ^2: unbiased.

Componentwise check. Each observation has the mean of its own group, so $\mathrm{E}\{\bar{Y}_1\} = a_1$ and $\mathrm{E}\{\bar{Y}_2\} = a_2$. Therefore in parameterization A

$$\boxed{\mathrm{E}\{\hat{a}_1\} = a_1, \qquad \mathrm{E}\{\hat{a}_2\} = a_2,}$$

and in parameterization B, by linearity of expectation,

$$\boxed{\mathrm{E}\{\hat{b}_0\} = \mathrm{E}\{\bar{Y}_1\} = b_0, \qquad \mathrm{E}\{\hat{b}_1\} = \mathrm{E}\{\bar{Y}_2\} - \mathrm{E}\{\bar{Y}_1\} = a_2 - a_1 = b_1.}$$

Unbiasedness needs only normality's first moment and the common-mean-within-group structure; the equal-variance and normality assumptions play no role here—they enter only when we turn to the *spread* of these estimators in Parts 7–11. Note this is exactly the linear-unbiasedness constraint of Assignment 9's Exercise 4 realized by a specific weighting: $\bar{Y}_g$ puts weight $1/n_g$ on each observation in its group. ■

Part 6

Provide a heuristic argument that the MLE for the common variance is biased.

Solution:

The variance MLE averages squared residuals *about the fitted means* rather than about the true means,

$$\hat{\sigma}^2 = \frac{1}{n}\Big[\sum_i (Y_{1i} - \bar{Y}_1)^2 + \sum_j (Y_{2j} - \bar{Y}_2)^2\Big].$$

The heuristic is that $\bar{Y}_g$ is pulled toward its own group's data—it is the very point that *minimizes* that group's sum of squared deviations—so residuals taken about $\bar{Y}_g$ are systematically smaller than the errors $Y_{gi} - \mu_g$ taken about the unknown true mean. We used the data twice: once to locate the center, again to measure spread about it, and the second measurement is optimistic.

Made quantitative, the shrinkage is exactly two observations' worth. For a single normal sample the standard identity $\mathrm{E}\big[\sum_{i=1}^m (Y_i - \bar{Y})^2\big] = (m-1)\sigma^2$ costs one degree of freedom for the one estimated mean; applying it to each group,

$$\mathrm{E}\{\mathrm{RSS}\} = \mathrm{E}\Big[\sum_i (Y_{1i} - \bar{Y}_1)^2\Big] + \mathrm{E}\Big[\sum_j (Y_{2j} - \bar{Y}_2)^2\Big] = (n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2 = (n-2)\sigma^2.$$

Two means were estimated, so two degrees of freedom are spent, and

$$\boxed{\mathrm{E}\{\hat{\sigma}^2\} = \frac{n-2}{n}\,\sigma^2 < \sigma^2,}$$

biased downward by the factor $(n-2)/n$. The bias shrinks to zero as $n \to \infty$ but never vanishes in finite samples; dividing the same RSS by $n-2$ instead of n gives the unbiased estimator $s^2 = \mathrm{RSS}/(n-2)$ used from Part 8 onward. In the matrix picture the reason the count is $n-2$ is geometric: the residual vector $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$ lives in the orthogonal complement of the two-dimensional column space of $\mathbf{X}$, a subspace of dimension $n-2$, so it can hold only $n-2$ error directions' worth of variance. ■

Part 7

Show that the regression coefficient estimates are independent of the estimate of the common variance. Use the facts that linear combinations of independent normals have a joint normal distribution, that jointly normal variables are independent if they have zero covariance, and that the coefficients and the variance estimator are functions of jointly normal vectors.

SOLUTION:

Both $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are built from two vectors that are *linear* in the Gaussian data $\mathbf{Y}$:

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \mathbf{A}\mathbf{Y}, & \mathbf{A} &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top, \\ \mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} &= (\mathbf{I} - \mathbf{H})\mathbf{Y}, & \mathbf{H} &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top, \end{aligned}$$

where $\mathbf{A}$ recovers the coefficients and $\mathbf{H}$ (the "hat" matrix) produces the fitted values, so that $\hat{\sigma}^2 = \mathbf{e}^\top\mathbf{e}/n$ is a function of the residual vector $\mathbf{e}$ alone. Because $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are linear images of the same multivariate-normal $\mathbf{Y}$, the stacked vector $(\hat{\boldsymbol{\beta}}, \mathbf{e})$ is *jointly normal* (linear combinations of independent normals are jointly normal). For jointly normal vectors, independence is equivalent to zero cross-covariance, so it suffices to compute

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \mathbf{A}\ \mathrm{Cov}(\mathbf{Y})\,(\mathbf{I} - \mathbf{H})^\top = \sigma^2\,\mathbf{A}(\mathbf{I} - \mathbf{H}),$$

using $\mathrm{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}_n$ and the symmetry of $\mathbf{I} - \mathbf{H}$. Now

$$\mathbf{A}(\mathbf{I} - \mathbf{H}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{I} - \mathbf{H}) = (\mathbf{X}^\top\mathbf{X})^{-1}\big(\mathbf{X}^\top - \mathbf{X}^\top\mathbf{H}\big),$$

and $\mathbf{X}^\top\mathbf{H} = \mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{X}^\top$, so $\mathbf{X}^\top - \mathbf{X}^\top\mathbf{H} = \mathbf{0}$. Therefore

$$\boxed{\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \mathbf{0} \implies \hat{\boldsymbol{\beta}} \perp \mathbf{e} \implies \hat{\boldsymbol{\beta}} \perp \hat{\sigma}^2.}$$

The last implication is because $\hat{\sigma}^2$ (indeed the whole of s^2) is a function of $\mathbf{e}$, and a function of a vector independent of $\hat{\boldsymbol{\beta}}$ is itself independent of $\hat{\boldsymbol{\beta}}$. The argument used precisely the three supplied facts: joint normality of linear images, the covariance-zero criterion for normal independence, and the representation of both estimators as functions of jointly normal vectors. Nothing here depended on the parameterization—$\hat{\boldsymbol{\beta}}$ changes basis with $\mathbf{X}$, but the residual vector $\mathbf{e}$, hence $\hat{\sigma}^2$, is the same object, orthogonal to the (common) column space either way. ■

Part 8

How would you compute standard errors for the regression coefficient estimates?

Solution:

The covariance matrix of the coefficient vector is

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{A}\ \mathrm{Cov}(\mathbf{Y})\,\mathbf{A}^\top = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

The variance of $\hat{\beta}_j$ is the j-th diagonal entry, $\sigma^2\big[(\mathbf{X}^\top\mathbf{X})^{-1}\big]_{jj}$. Since σ^2 is unknown, plug in the unbiased $s^2 = \mathrm{RSS}/(n-2)$ of Part 6; the standard error is the square root of the estimated variance:

$$\boxed{\widehat{\mathrm{SE}}(\hat{\beta}_j) = \sqrt{s^2\big[(\mathbf{X}^\top\mathbf{X})^{-1}\big]_{jj}}, \qquad s^2 = \frac{1}{n-2}\Big[\sum_i (Y_{1i}-\bar{Y}_1)^2 + \sum_j (Y_{2j}-\bar{Y}_2)^2\Big].}$$

Reading off the diagonals computed in Part 4 makes this concrete. In parameterization A, $(\mathbf{X}_A^\top\mathbf{X}_A)^{-1} = \mathrm{diag}(1/n_1, 1/n_2)$, so

$$\widehat{\mathrm{SE}}(\hat{a}_1) = \frac{s}{\sqrt{n_1}}, \qquad \widehat{\mathrm{SE}}(\hat{a}_2) = \frac{s}{\sqrt{n_2}},$$

the familiar standard error of a group mean. In parameterization B, $(\mathbf{X}_B^\top\mathbf{X}_B)^{-1} = \frac{1}{n_1 n_2}\left(\begin{smallmatrix} n_2 & -n_2 \\ -n_2 & n \end{smallmatrix}\right)$ has diagonal entries $n_2/(n_1 n_2) = 1/n_1$ and $n/(n_1 n_2) = 1/n_1 + 1/n_2$, so

$$\widehat{\mathrm{SE}}(\hat{b}_0) = \frac{s}{\sqrt{n_1}}, \qquad \widehat{\mathrm{SE}}(\hat{b}_1) = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

The intercept's standard error matches $\hat{a}_1$'s, since $\hat{b}_0 = \hat{a}_1$, while $\hat{b}_1$'s is the classical two-sample standard error for a difference of means: estimating a contrast pays the variance of *both* group means. The off-diagonal $-n_2/(n_1 n_2) = -1/n_1$ records that $\hat{b}_0$ and $\hat{b}_1$ are negatively correlated in this coding, whereas $\hat{a}_1, \hat{a}_2$ are uncorrelated. $\blacksquare$

Part 9

Would the standard errors be independent of the regression coefficients?

Solution:

Yes. A standard error $\widehat{\mathrm{SE}}(\hat{\beta}_j) = \sqrt{s^2\,[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}}$ is a deterministic constant times s, and $s^2 = \mathrm{RSS}/(n-2) = \mathbf{e}^\top\mathbf{e}/(n-2)$ is a function of the residual vector $\mathbf{e}$ alone. Part 7 established that $\mathbf{e}$ is independent of $\hat{\boldsymbol{\beta}}$; hence any function of $\mathbf{e}$—in particular each standard error—is independent of the coefficient estimates:

$$\boxed{\widehat{\mathrm{SE}}(\hat{\beta}_j) \text{ is a function of } \mathbf{e}, \quad \mathbf{e} \perp \hat{\boldsymbol{\beta}} \implies \widehat{\mathrm{SE}}(\hat{\beta}_j) \perp \hat{\boldsymbol{\beta}}.}$$

This is what lets the t statistic of Part 11 have a clean distribution: dividing $\hat{\beta}_j$ (or a contrast) by its estimated standard error puts an independent numerator over an independent denominator, which is exactly the standard-normal-over-chi structure that defines Student's t. ■

Part 10

How would you compute standard errors for linear combinations of the regression coefficient estimates?

Solution:

Let $\mathbf{c}$ be a fixed 2×1 vector and consider the scalar $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$. Its variance follows from $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$ by the quadratic-form rule $\mathrm{Var}(\mathbf{c}^\top\hat{\boldsymbol{\beta}}) = \mathbf{c}^\top \mathrm{Cov}(\hat{\boldsymbol{\beta}})\mathbf{c}$:

$$\mathrm{Var}(\mathbf{c}^\top\hat{\boldsymbol{\beta}}) = \sigma^2\, \mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}, \qquad \text{so} \qquad \boxed{\widehat{\mathrm{SE}}(\mathbf{c}^\top\hat{\boldsymbol{\beta}}) = s\sqrt{\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}}\,.}$$

This single formula subsumes Part 8 (take $\mathbf{c} = e_j$, a standard basis vector, to recover the j-th coefficient) and delivers the standard error of any contrast. It also lets the two codings be cross-checked: the treatment effect $a_2 - a_1$ is $\mathbf{c}^\top(a_1, a_2)^\top$ with $\mathbf{c} = (-1, 1)^\top$ in parameterization A, whose standard error is

$$s\sqrt{(-1,1)\,\mathrm{diag}(1/n_1, 1/n_2)(-1,1)^\top} = s\sqrt{\tfrac{1}{n_1} + \tfrac{1}{n_2}},$$

identical to $\widehat{\mathrm{SE}}(\hat{b}_1)$ from Part 8, because $a_2 - a_1$ and b_1 are the same random variable. The crucial point for a difference of correlated estimators is that one must use the full matrix $(\mathbf{X}^\top\mathbf{X})^{-1}$, including its off-diagonal: $\mathrm{Var}(\hat{a}_2 - \hat{a}_1) = \mathrm{Var}\,\hat{a}_1 + \mathrm{Var}\,\hat{a}_2 - 2\,\mathrm{Cov}(\hat{a}_1, \hat{a}_2)$, and here the covariance term happens to vanish in coding A but not in coding B—yet both give the same answer, as they must. ■

Part 11

What would be the distribution of the ratio of a linear combination of regression coefficient estimates to its standard error? Show this through a derivation.

Solution:

Fix $\mathbf{c}$ and consider the studentized ratio, centered at the true value $\mathbf{c}^\top\boldsymbol{\beta}$,

$$T = \frac{\mathbf{c}^\top\hat{\boldsymbol{\beta}} - \mathbf{c}^\top\boldsymbol{\beta}}{\widehat{\mathrm{SE}}(\mathbf{c}^\top\hat{\boldsymbol{\beta}})} = \frac{\mathbf{c}^\top\hat{\boldsymbol{\beta}} - \mathbf{c}^\top\boldsymbol{\beta}}{s\sqrt{\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}}}.$$

We assemble it from a standard normal and an independent chi-square in three steps.

Step 1: a standard normal numerator. Since $\hat{\boldsymbol{\beta}}$ is normal with mean $\boldsymbol{\beta}$ and covariance $\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$, the scalar $\mathbf{c}^\top\hat{\boldsymbol{\beta}}$ is normal with mean $\mathbf{c}^\top\boldsymbol{\beta}$ and variance $\sigma^2\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}$. Standardizing with the *true* σ,

$$Z = \frac{\mathbf{c}^\top\hat{\boldsymbol{\beta}} - \mathbf{c}^\top\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}}} \sim N(0,1).$$

Step 2: an independent chi-square. With $\mathbf{e} = (\mathbf{I}-\mathbf{H})\mathbf{Y}$ and $\mathbf{I}-\mathbf{H}$ an idempotent projection of rank $n-2$,

$$W = \frac{(n-2)s^2}{\sigma^2} = \frac{\mathrm{RSS}}{\sigma^2} = \frac{\mathbf{e}^\top\mathbf{e}}{\sigma^2} \sim \chi^2_{n-2}.$$

(The residual sum of squares of a normal linear model, divided by σ^2, is chi-square with degrees of freedom equal to the rank of the residual projection, here $n-2$.)

Step 3: independence and assembly. By Part 7, $\hat{\boldsymbol{\beta}} \perp \mathbf{e}$, so $Z \perp W$. The definition of Student's t as a standard normal over the square root of an independent chi-square divided by its degrees of freedom gives

$$\frac{Z}{\sqrt{W/(n-2)}} = \frac{\big(\mathbf{c}^\top\hat{\boldsymbol\beta} - \mathbf{c}^\top\boldsymbol\beta\big)\big/\big(\sigma\sqrt{\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}}\big)}{\sqrt{\big[(n-2)s^2/\sigma^2\big]/(n-2)}} = \frac{\mathbf{c}^\top\hat{\boldsymbol\beta} - \mathbf{c}^\top\boldsymbol\beta}{s\sqrt{\mathbf{c}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{c}}} = T,$$

the two factors of σ cancelling. Therefore

$$\boxed{T = \frac{\mathbf{c}^\top\hat{\boldsymbol\beta} - \mathbf{c}^\top\boldsymbol\beta}{\widehat{\mathrm{SE}}(\mathbf{c}^\top\hat{\boldsymbol\beta})} \;\sim\; t_{n-2}.}$$

The degrees of freedom $n-2$ are the two the variance estimate "lost" in Part 6 to the two group means; replacing the unknown σ by the estimate s is what fattens the tails from normal to t. Taking $\mathbf{c} = (-1,1)^\top$ in parameterization A (equivalently $\mathbf{c} = (0,1)^\top$ in parameterization B) makes T the two-sample t statistic for the treatment effect $b_1 = a_2 - a_1$. So the pooled-variance two-sample t test is just this contrast in the two-group linear model. The answer does not depend on the coding, since $\mathbf{c}^\top\hat{\boldsymbol\beta}$, its standard error, and the residuals are the same objects in both. ■

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